

A Regression Analysis to Predict Life Expectancy Based on Demographic and Socioeconomic Indicators

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Abstract— One of the most element in end of life judgement is Life expectancy. This research explores the computation of machine learning regression techniques to predict life expectancy using a dataset of 15 years collected from Kaggle, comprising demographic, health, and socioeconomic features. The dataset contain information from 193 countries with multiple variables such as 'Country', 'Year', 'Status' (Developer/Developing), and many numerical indicators (e.g., Adult Mortality, GDP, and Schooling). Twice Exploratory Data Analysis (EDA) was performed to gain proper insights into feature distributions, identify outliers, and assess correlations. Missing values were treated using both mean imputation and kNN imputation([6]), and categorical variables were encoded by label encoding. A rapid model benchmarking approach is applied using the LazyPredict library was employed to compare multiple regression models, from where many models were selected for Hyperparameter tuning to enhance their performance. The Regression model which achieve outstanding performance after doing Hyperparameter Tunning on the test set is XGBRegressor model([4], [5]) also the R2 score of model is 96.87% with minimal RMSE. These results highlight the potential and challenges of predictive modeling in life expectancy estimation, and underscore the importance of thorough preprocessing and model validation.

Keywords— *Life Expectancy Prediction, Machine Learning, Regression Models.*

1. INTRODUCTION

Life expectancy is an important measure of the health and wellness of the nation. Prediction of life expectancy may assist policymakers and healthcare professionals to formulate improved intervention strategies and anticipate resource allocation. This research predicts life expectancy based on a range of heterogeneous traits such as mortality rate, health expenditure, coverage of immunization, and economic factors.

Traditional statistical approaches are likely to miss the complex, non-linear relationships that occur in such data, but machine learning is a good alternative (as demonstrated in [1], [3], [4]). In this project, we employ regression techniques to predict life expectancy from a broad array of features, including everything from health measures (e.g., adult mortality, infant mortality) to economic indicators (e.g., GDP, schooling). Our objectives are to preprocess the data effectively, compare different regression models, and interpret the findings to identify key predictors of life expectancy. This work demonstrates the potential of machine learning in addressing real-world healthcare issues ([3], [5], [9], [10]).

2. Literature review

Prediction of life expectancy is increasingly becoming a fast-growing area of research in healthcare analytics and policy-making. Following recent advances in machine learning, new avenues have opened up to develop predictive models that can estimate complex relationships between demographic, socioeconomic, and health-based factors to precisely forecast life expectancy. Many researchers have investigated the application of machine learning techniques in forecasting life expectancy. Kumar and S. Gupta conducted a comparative analysis of regression algorithms in predicting life expectancy, during which they recognized the capability of these algorithms for identifying and describe intricate relationships that may not be detected using traditional statistical analysis [3]. Their study demonstrated the different regression models' performance in dealing with health indicators and relating

them to life expectancy outcomes. Similarly, Lakshmanarao et al. investigated the interplay between immunization coverage and Human Development Index (HDI) indicators in influencing life expectancy [1]. Their work focused particularly on how preventive health interventions greatly contribute to population longevity when coupled with socioeconomic development factors. This multi-factor model has been instrumental in creating more refined predictive models.

Identification of influential predictors has been a major focus area in life expectancy modeling. Vydehi et al. investigated the importance of immunization indicators along with HDI indicators in their research, confirming that health interventions and socioeconomic status are primary determinants of population health outcomes [5]. Their research highlights the importance of proper feature selection while building predictive models. Based on this understanding, Adebayo and Adeyemo employed supervised machine learning models to predict life expectancy, with emphasis on ranking feature importance [4]. From their research, they established strong variables such as adult mortality, schooling, and income composition to be most closely related to life expectancy, as per other studies in this field. Besides general predictive modeling, specific applications have been seen in the literature. Nishi constructed a prediction model of healthy life years with no activity limitation, an extension of the prediction of life expectancy to incorporate quality-of-life considerations [8]. They presented the adaptability of machine learning techniques to address various facets of population health prediction.

S.K. Singh and R. Sharma compared decision tree classification and linear regression algorithms for predicting life expectancy, gaining an insight into the comparative advantages of different algorithmic methods [6]. It was indicated from their study that tree-based models had the potential to identify non-linear trends that were not identifiable in linear models, particularly when dealing with demographic and health variables. Researchers have also tackled the issue of working with international health datasets. Dinov's comprehensive review of biomedical and health applications highlighted the importance of proper data preprocessing, including missing value management, outlier detection, and feature scaling [7]. The preprocessing has been shown to be required in developing robust predictive models, especially when dealing with disparate international datasets with heterogeneous health indicators. Life expectancy forecasting literature based on machine learning has demonstrated significant advances over the past few decades. Researchers have moved from simple linear models to sophisticated ensemble techniques while at the same time expanding their scope to relevant predictor variables.

3. DATASET DESCRIPTION

The Global Health Observatory (GHO) data repository under World Health Organization (WHO) keeps track of the health status as well as many other related factors for all countries. The datasets are made available to public for the purpose of health data analysis and I take the dataset from **opensource Kaggle**[2]. The dataset related to life expectancy, health factors for 193 countries has been collected from the same WHO data repository website ([2]) and its corresponding economic data was collected from United Nation website. Among all categories of health-related factors only those critical factors were chosen which are more representative. It has been observed that in the past 15 years, there has been a huge development in health sector resulting in improvement of human mortality rates especially in the developing nations in comparison to the past 30 years. Therefore, in this project we have considered data from year 2000-2015 for 193 countries for further analysis. The dataset, sourced from the **World Health Organization (WHO)** and **United Nations (UN)**, includes **demographic, economic, and health-related factors** that may impact life expectancy. By leveraging statistical and machine learning models, I aim to identify key predictors affecting life expectancy and provide insights that can help countries improve their healthcare strategies.

The dataset consists of **2938 records** with **22 columns**, categorized as follows:

Demographic Factors

- **Country:** Name of the country
- **Year:** The year the data was recorded
- **Status:** Indicates whether the country is Developed or Developing

Health Factors

- **Life expectancy:** Expected number of years a newborn is likely to live (*Target Variable*)
- **Adult Mortality:** Mortality rate of adults per 1000 population
- **Infant deaths:** Number of infant deaths per 1000 births

- **Under-five deaths:** Number of deaths of children under five per 1000 births
- **Hepatitis B:** Immunization rate of Hepatitis B
- **Measles:** Number of reported measles cases
- **Polio:** Immunization rate for Polio
- **Diphtheria:** Immunization rate for Diphtheria
- **HIV/AIDS:** Death rate per 1000 people due to HIV/AIDS
- **BMI:** Average Body Mass Index of the population
- **Thinness 1-19 years:** Prevalence of thinness among children and adolescents (1-19 years)
- **Thinness 5-9 years:** Prevalence of thinness among children (5-9 years)

Economic Factors

- **GDP:** Gross Domestic Product per capita
- **Schooling:** Average number of years of schooling
- **Income composition of resources:** Human Development Index income component
- **Percentage expenditure:** Total health expenditure as a percentage of GDP
- **Total expenditure:** Government expenditure on health as a percentage of total expenditure

Population and Resources Factors

- **Population:** Total population of the country

The dataset comprises observations from 193 countries across multiple years, featuring 21 variables, with '**Life Expectancy**' as the target variable. The features are:

- **Categorical:**
 - 'Country' (193 unique values)
 - 'Status' (Developed or Developing)
- **Numerical:**
 - 'Year', 'Adult Mortality', 'Infant Deaths', 'Alcohol', 'Percentage Expenditure', 'Hepatitis B', 'Measles', 'BMI', 'Under-Five Deaths', 'Polio', 'Total Expenditure', 'Diphtheria', 'HIV/AIDS', 'GDP', 'Population', 'Thinness 1-19 Years', 'Thinness 5-9 Years', 'Income Composition of Resources', 'Schooling'.

4. Proposed Methodology

Figure 1 shows the framework of this Research Paper. Initially, We collected dataset from Kaggle [2] with multiple features. All the features were preprocessed and visualized to find best insights with graphs. After that Lazy Predict library is applied and looking on the result we choose best 9 Regression Models for Hyperparameter Tunning to improve their accuracy. After that, the algorithm with highest accuracy is chosen to predict Life expectancy

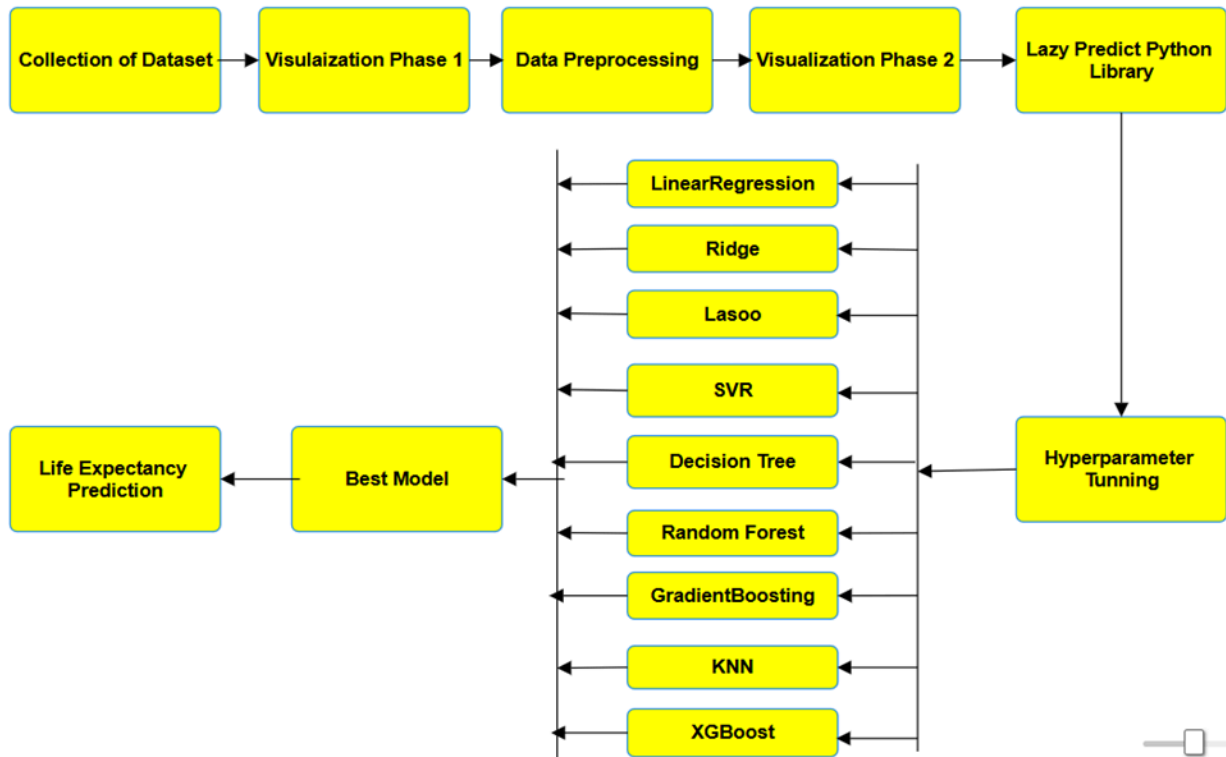


Fig.1. Proposed Method

5. Exploratory Data Analysis

The dataset has been subjected to rigorous preprocessing procedures to improve the quality of results produced in the research. This can be by performing initial EDA that includes all associated sanity checks. Preprocessing also include the computation of the descriptive statistics from the acquired dataset to understand the distribution of numerical variables present in the dataset. Dataset is further filtered with the exploratory data analysis in detail: which is carried out in two phases viz. with and without preprocessing. The detailed analysis of the dataset indicated that most of the missing data was for population, Hepatitis B and GDP. The missing data were from less known countries like Vanuatu, Tonga, Togo, Cabo Verde etc. Finding all data for these countries was difficult and hence, it was decided that we exclude these countries from the final model dataset. The final merged file(final dataset) consists of 22 Columns and 2938 rows which meant 20 predicting variables. All predicting variables were then divided into several broad categories: Immunization related factors, Mortality factors, Economical factors and Social factors. Initial inspection revealed missing values in several columns, necessitating preprocessing, but no duplicate rows were found. The dataset's diversity—spanning health, economic, and demographic domains—makes it suitable for regression analysis, though the high cardinality of 'Country' poses encoding challenges. To understand the shape and distribution attributes in the dataset EDA has been plotted in phases. First Phase without doing any preprocessing and second Phase after doing preprocessing

EDA First Phase:- the detailed description of the working of First phase has been illustrated in Fig.2 below. Where **Histograms** has been Plotting Histogram for all features (**Univariate Analysis**) to get basic overview of data:-

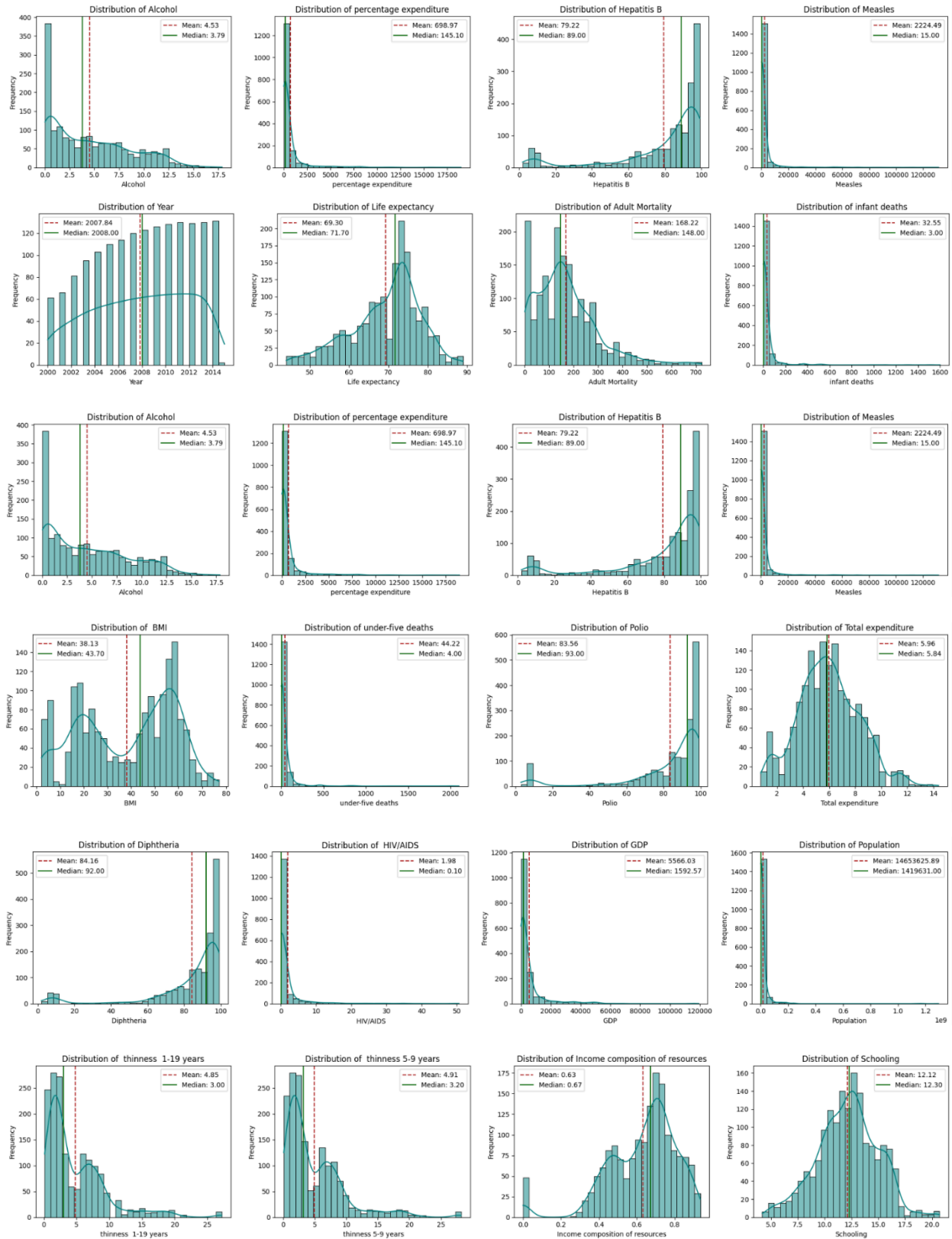


Fig. 2. Univariate Analysis of all columns

Insights from histogram analysis illustrated above in Fig.1.:

1. It has been observed that '**Adult Mortality**' and '**Alcohol**' exhibited a left skew (see. Fig 3,4 respectively).

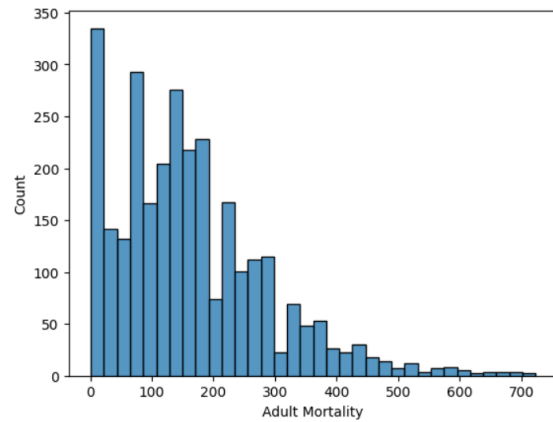


Figure 3. Adult Mortality Count

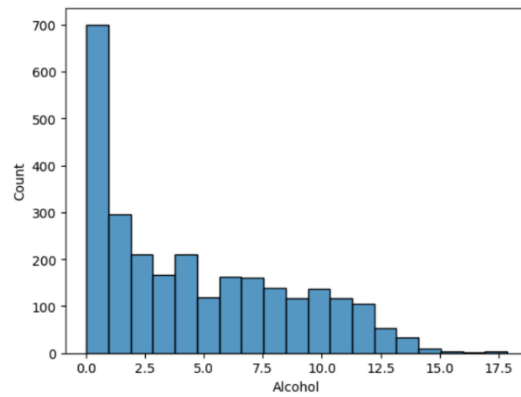


Figure 4. Alcohol consumption count

2. It has been observed that '**Schooling**' and '**Life expectancy**' exhibited a normal skew (see. Fig 5,6 respectively).

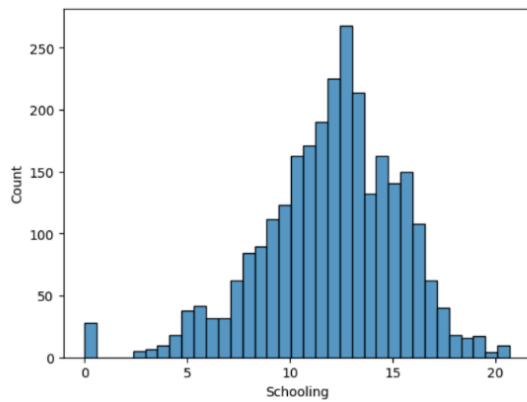


Figure 5. Schooling Count

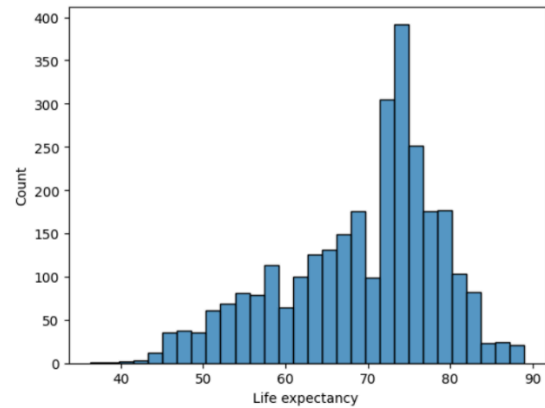


Figure 6. Alcohol life expectancy count

3. It has been observed that '**Polio**' and '**Income composition of resources**' exhibited a positive skew (see. Fig,7,8 respectively).

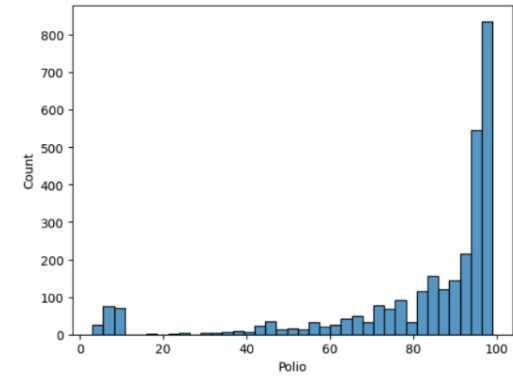


Figure 7. Polio Count

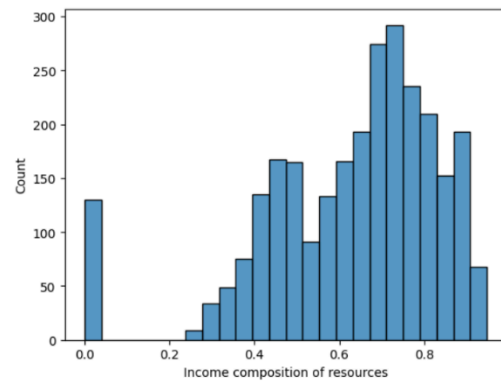


Figure 8. Income composition of resources count

To further analysis of the relationship of each accounted variable with the target variable, they are subjected to Bivariate Analysis. The results for the same can be visualized in the form of scatter plots, which has been created for each numerical feature against 'Life Expectancy' , for identifying strong relationships.

- 1) It has been observed that Negative correlation of Life Expectancy with '**Adult Mortality**' and '**HIV/AIDS**' shown in Figure 9.

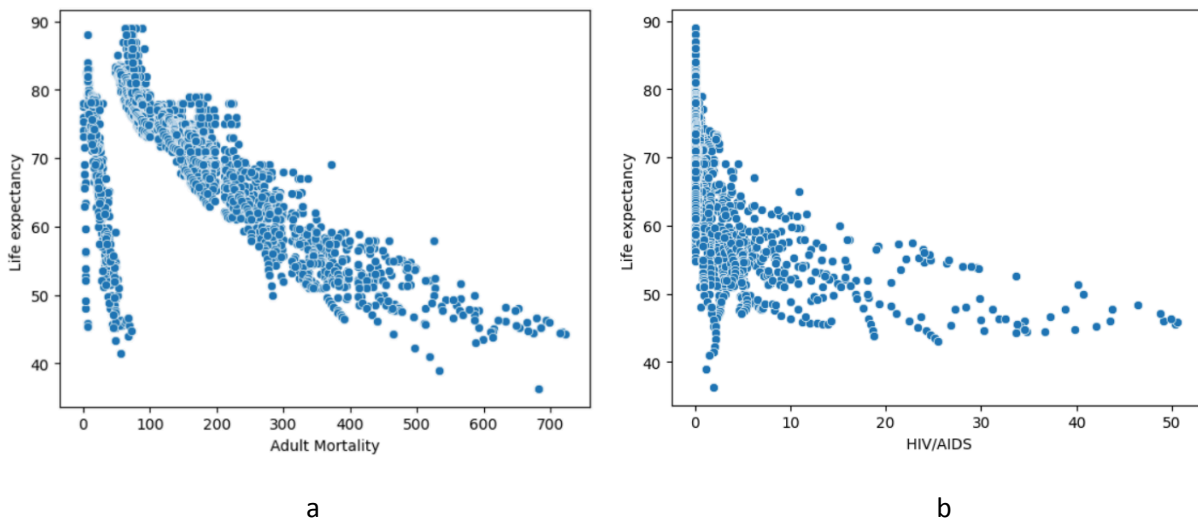


Fig. 9. a) Life Expectancy vs Adult Morality, b) Life Expectancy vs HIV/AIDS

- 2) It has been observed that Positive correlation of Life Expectancy with '**BMI**' and '**Schooling**' shown in Figure 10.

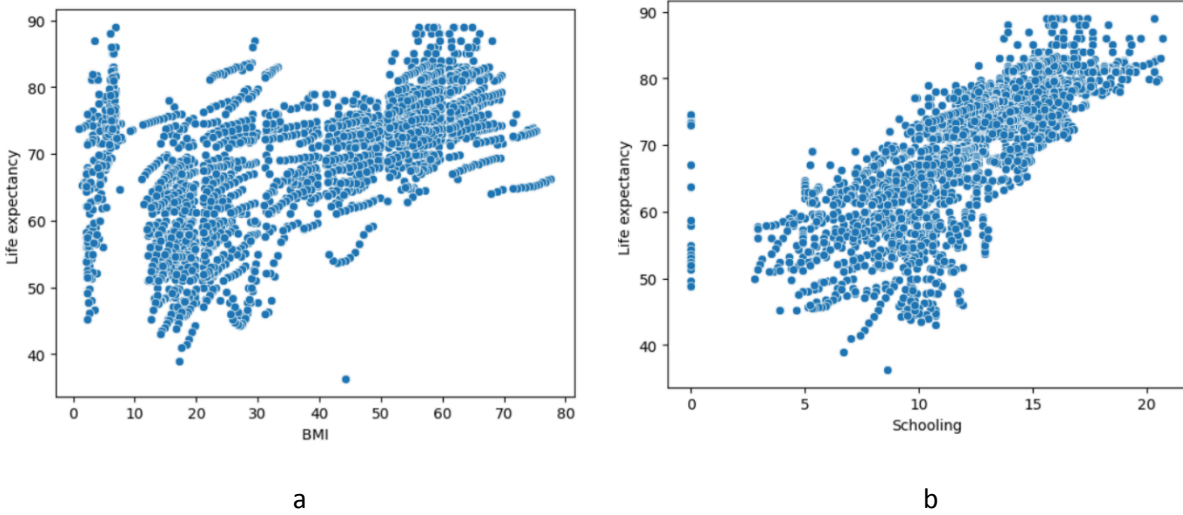


Fig. 10. a)Life Expectancy vs BMI, b) Life Expectancy vs schooling

To further understand the relationship of each variable with each other and other accounting variables they are subjected to Multivariate Analysis done using pearson correlation: A heatmap was drawn to understand the interrelationships among variables shown in Figure 11. A Notable correlations included 'Adult Mortality' (-0.70), 'Schooling' (0.75), and 'Income Composition of Resources' (0.72) with 'Life Expectancy'([1], [6], [8]). Features such as GDP, Schooling rates were strong indicators influencing life expectancy [3], [7], [8]

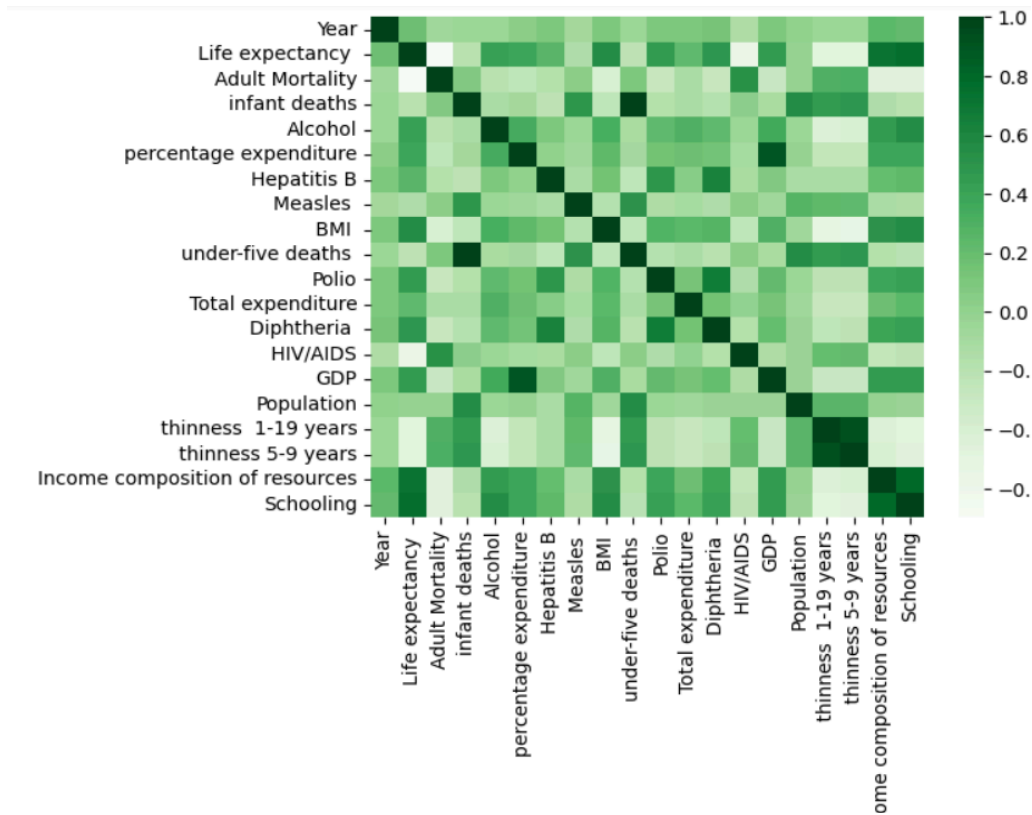


Fig. 11. Correlation Heatmap

6. Data Preprocessing:

6.1 Missing Value Imputation

We will perform missing values imputation which is done by majorly 2 methods Mean Imputation and KNN Imputer. **Mean Imputation** is Applied to numerical columns for a basic, fast fill-in of missing values using the column's mean. While simple, it can distort data variance and underrepresent feature relationships. **kNN Imputer** is a more advanced technique that imputes missing values by finding the most similar instances (nearest neighbors) based on other feature values. This method respects the underlying data distribution and relationships between features. kNN imputation was preferred over mean imputation due to its ability to preserve the natural structure and patterns of the data. Unlike mean imputation, which applies a global statistic, kNN adapts to local data neighborhoods—leading to more realistic and context-aware imputations. After that we have no missing value in our dataset.

6.2 Outlier Treatment:

- Firstly, we will identify outliers in every feature of our datasets. Outliers were identified by plotting boxplots showing only major outliers in (Fig. 12-19) and after then treated to reduce their impact on model training.

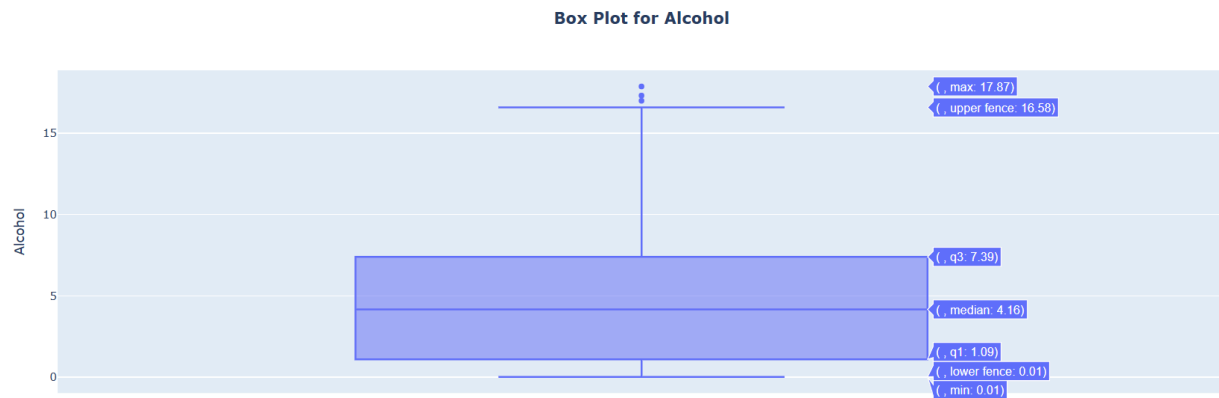


Fig. 12. Outlier in Alcohol

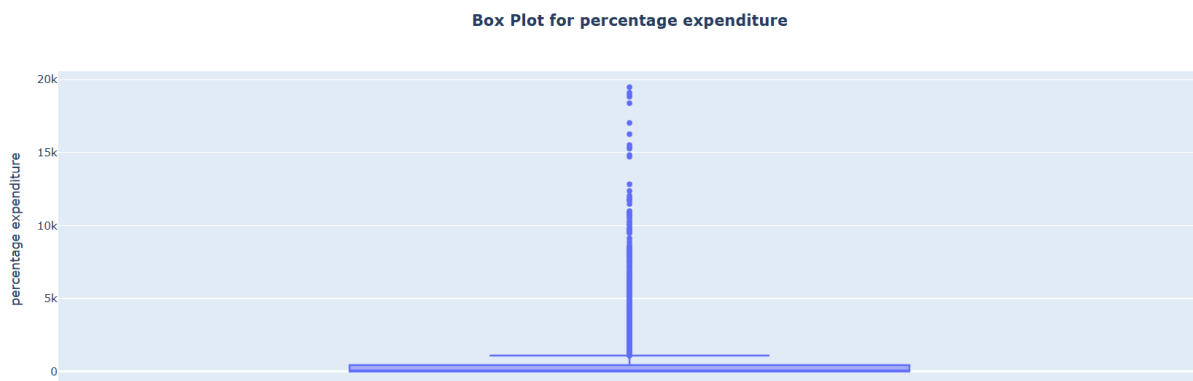


Fig. 13. Outlier in Percentage expenditure

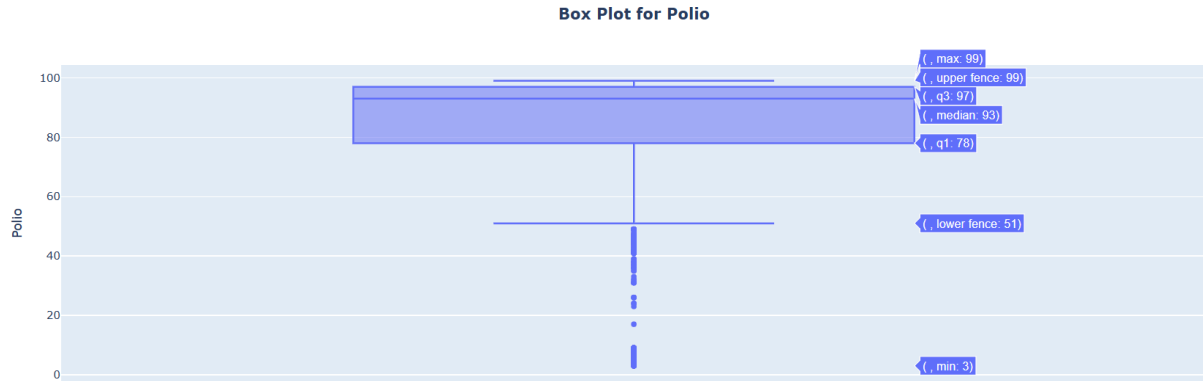


Fig. 14. Outlier in Polio

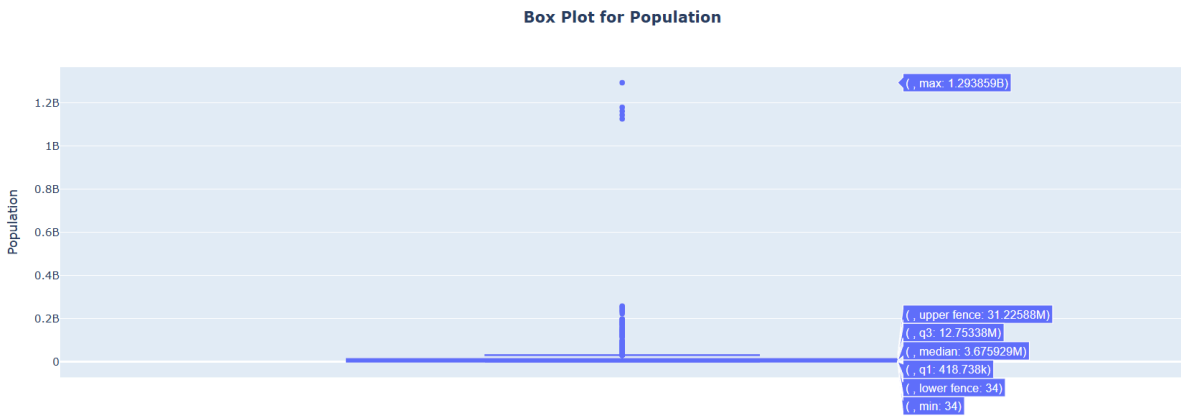


Fig. 15. Outlier in Population

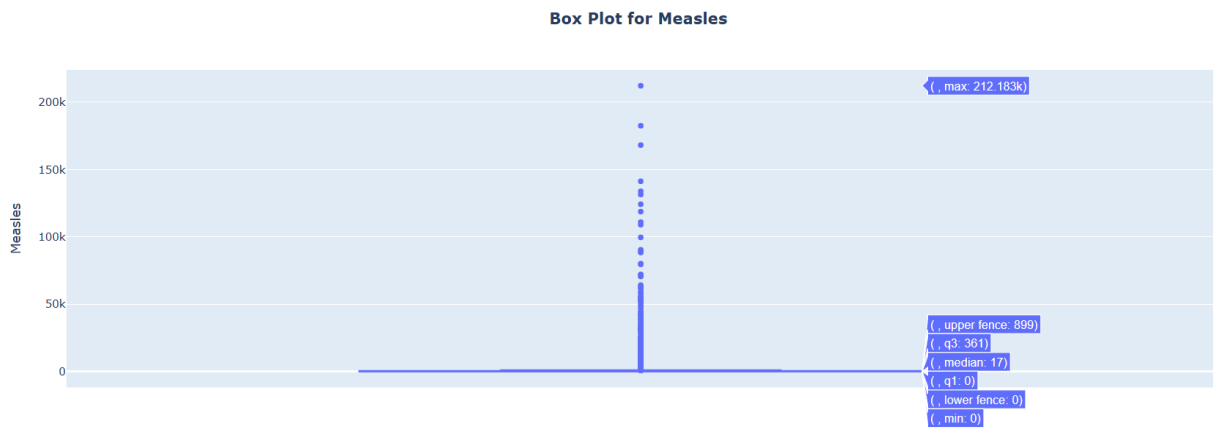


Fig. 16. Outlier in Measles

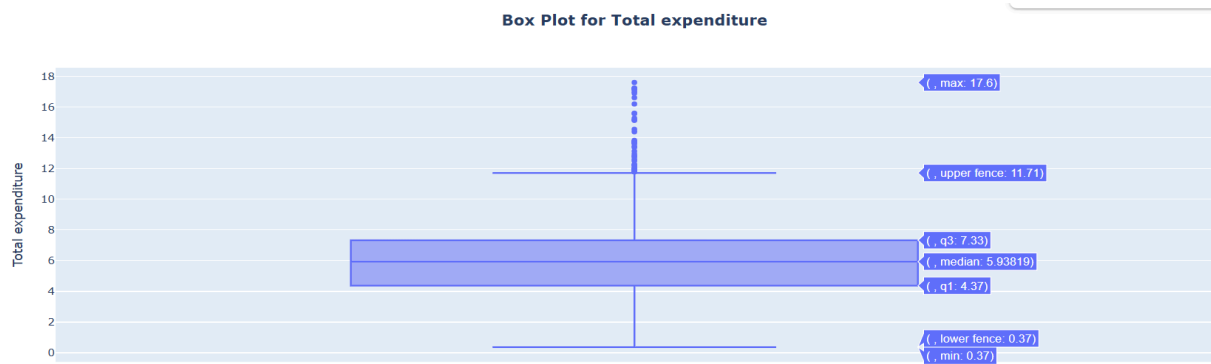


Fig. 17. Outlier in Total expenditure

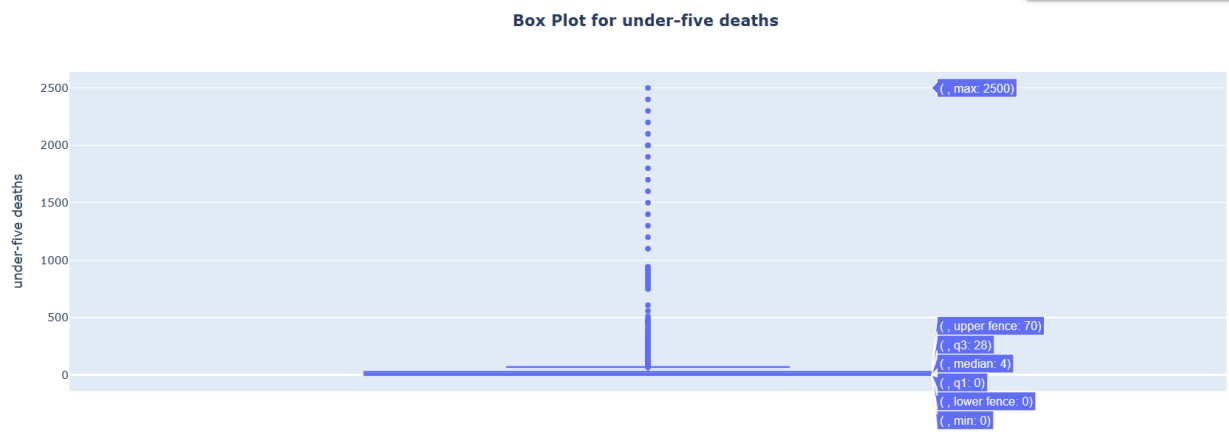


Fig. 18. Outlier in under-five deaths

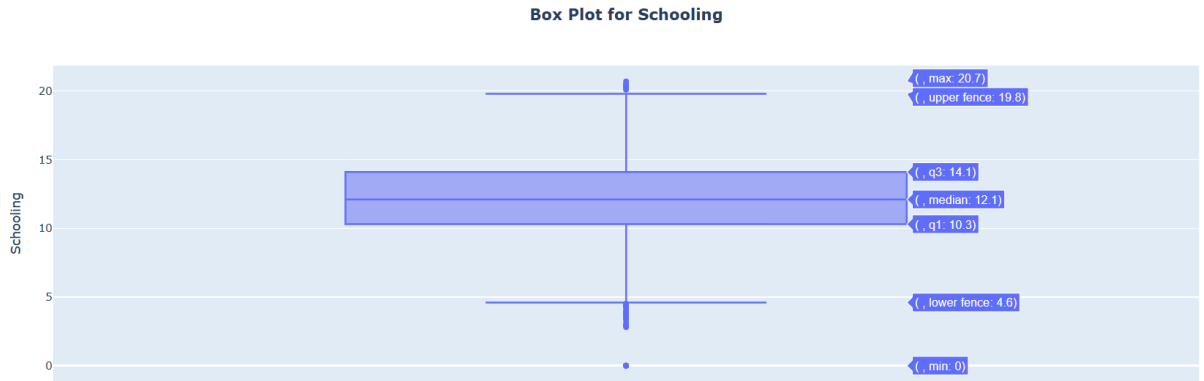


Fig. 19. Outlier in Schooling

Secondly, these outliers has been removed utilizing conventional detection treatments viz. IQR (Interquartile Range), which uses measure variability by dividing a data set into quartiles. The data is sorted in ascending order and then we split it into 4 equal parts. The values Q1 (25th percentile), Q2 (50th percentile or median) and Q3 (75th percentile) separate dataset in 4 equal parts.

- If a dataset has $2n$ or $2n+1$ data points, then
- $Q2$ = median of the dataset.
- $Q1$ = median of n smallest data points.
- $Q3$ = median of n highest data points.
- The IQR is calculated as: $IQR = Q3 - Q1 = Q3 - Q1$
- Data points that fall below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are considered outliers. Showing the graph of boxplot after Outliers Treatment. Also there are some graph like (“Adult- Mortality”, “Measles”, “Polio”, “Total expenditure” and “Schooling”) in which

After looking into the visualization shown in (Fig 20-32) obtained after the treatment of outliers **it has been observed from boxplot statistic measures that** Outliers are heavily reduced but not completely removed as data should contain some diversity which can help us to achieve **Low Bias and Low Variance** in model Training.

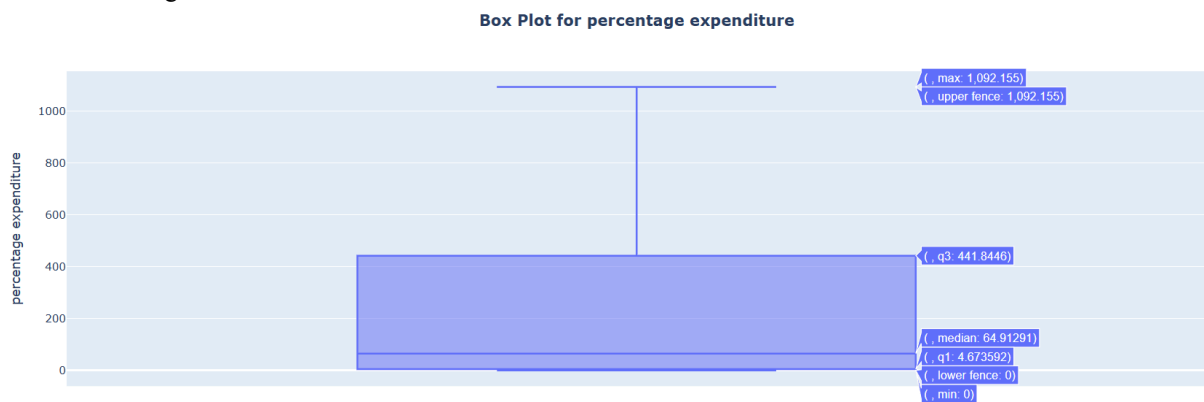


Fig. 20. Outlier removed from percentage expenditure

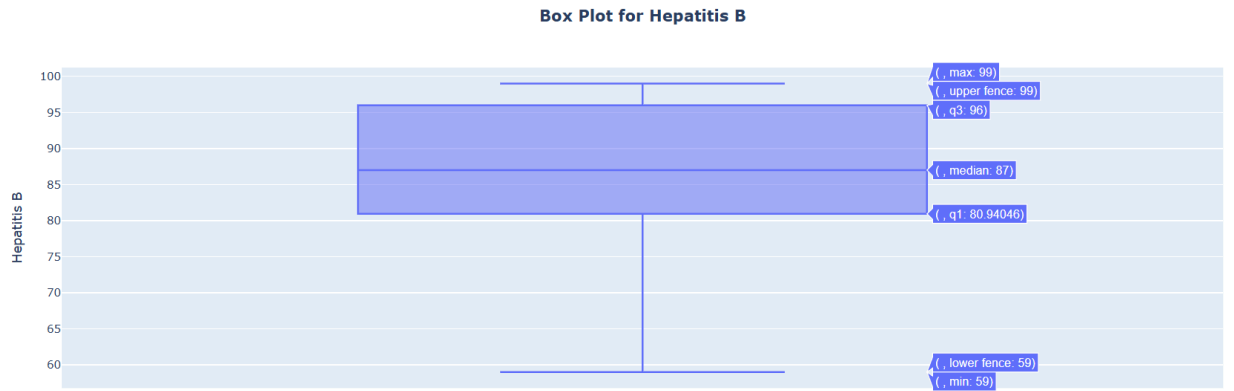


Fig. 21. Outlier removed from Hepatitis B

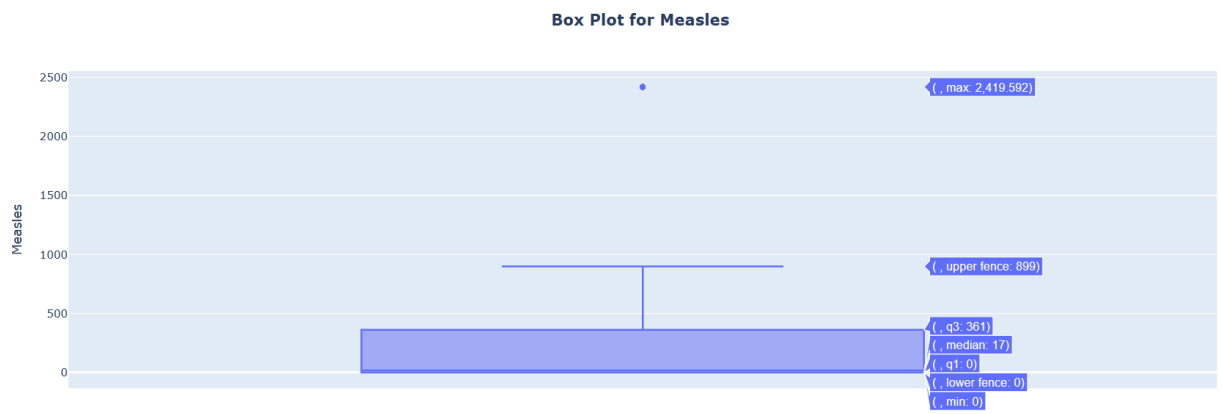


Fig. 22. One outlier left in Measles

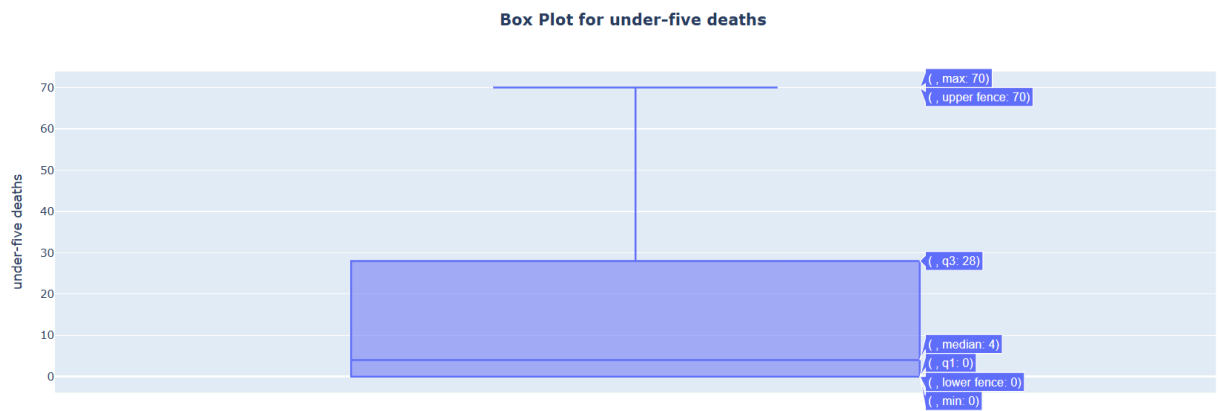


Fig. 23. Outlier removed from under-five deaths

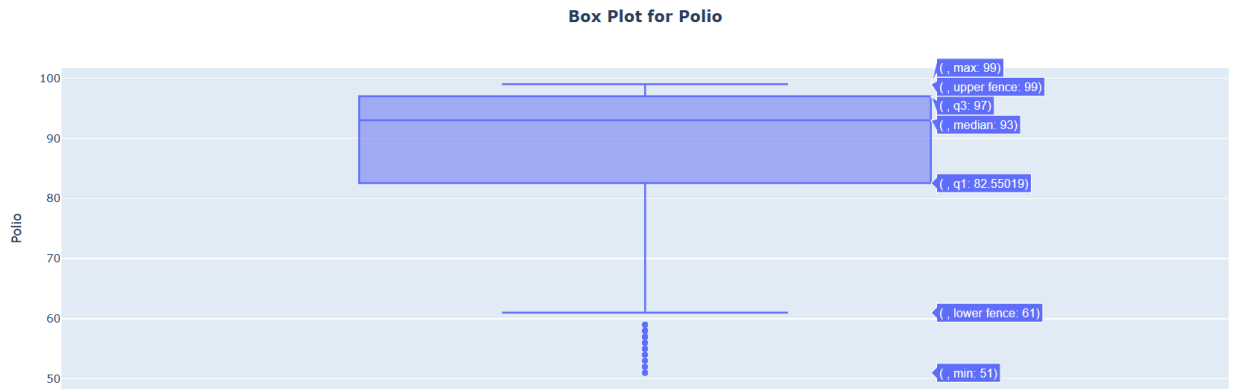


Fig. 24. Outlier heavily reduced from percentage expenditure

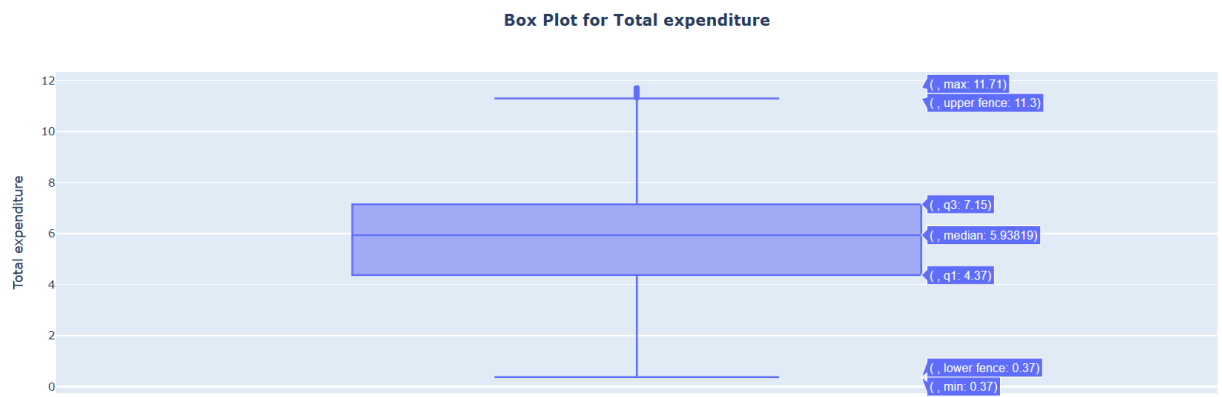


Fig. 25. Outlier heavily reduced from percentage expenditure

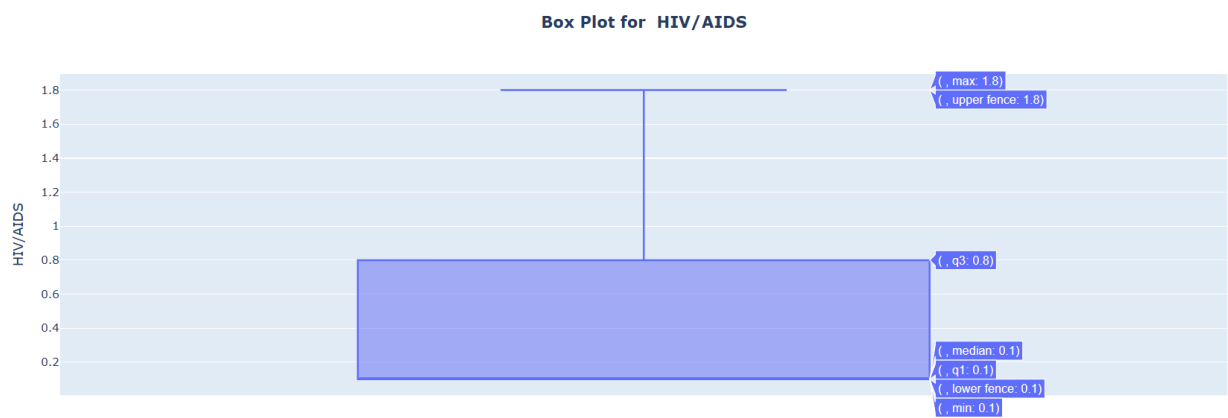


Fig. 26. Outlier removed from HIV/AIDS

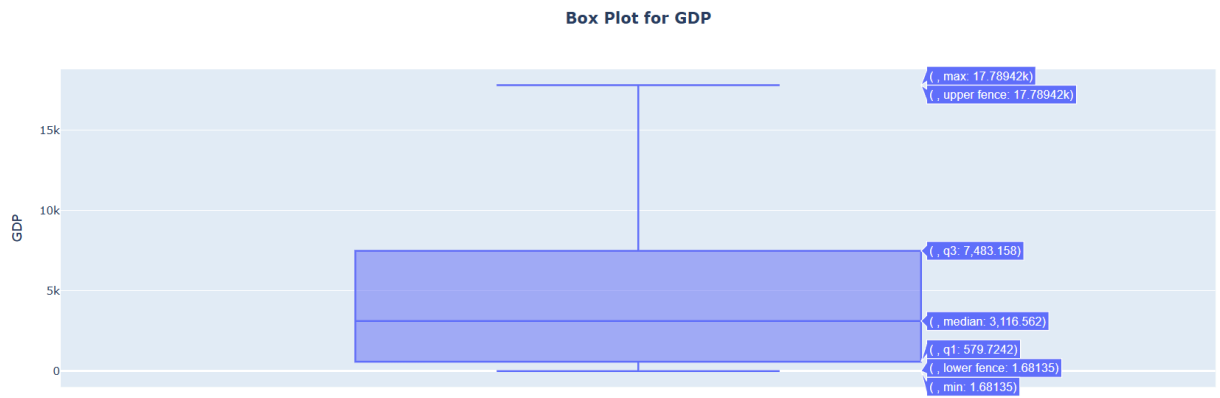


Fig. 27. Outlier removed from GDP

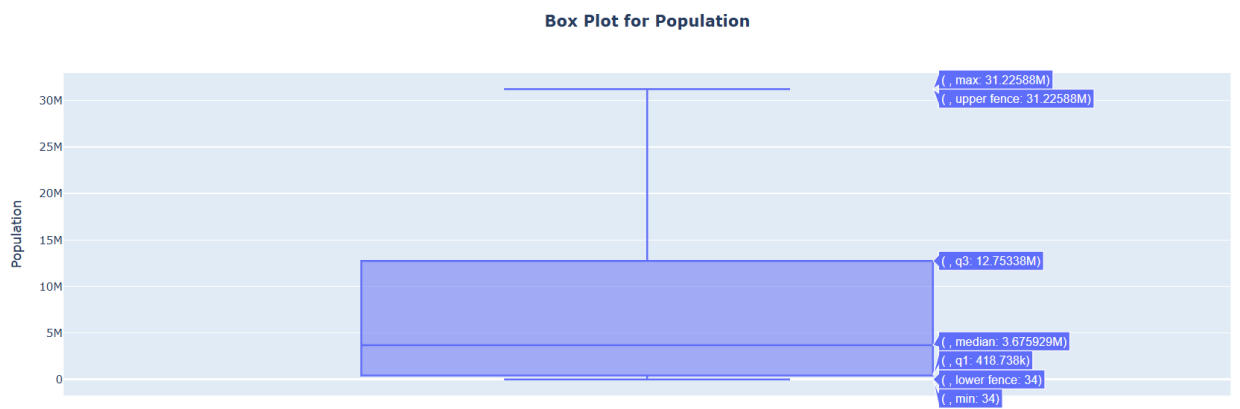


Fig. 28. Outlier removed from Population

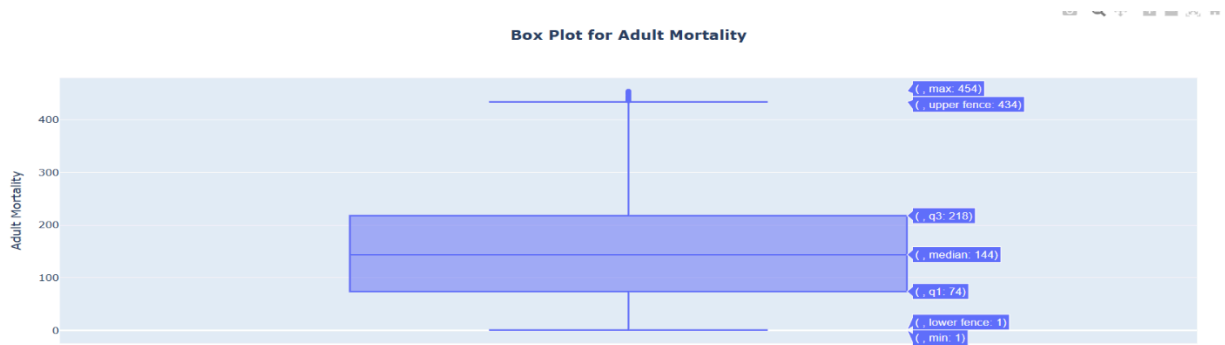


Fig. 29. Outlier heavily reduced from Adult Mortality

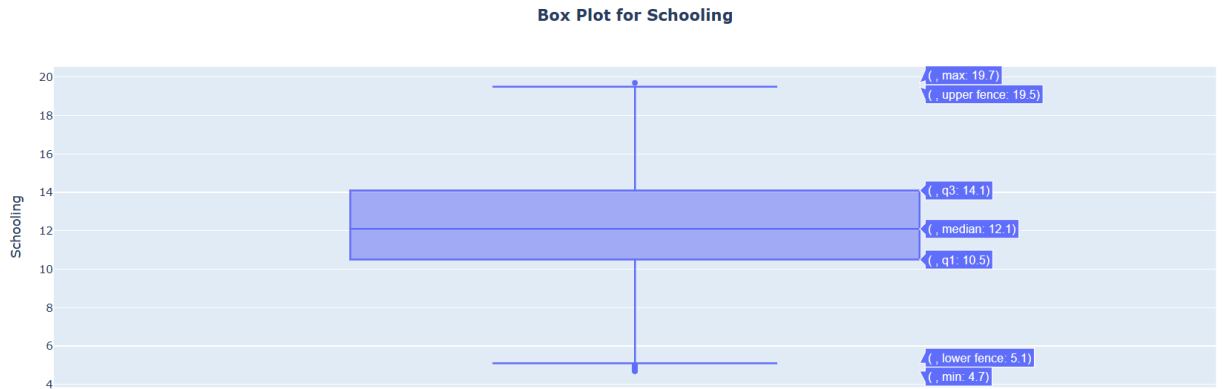


Fig. 30. Outlier heavily reduced from Schooling

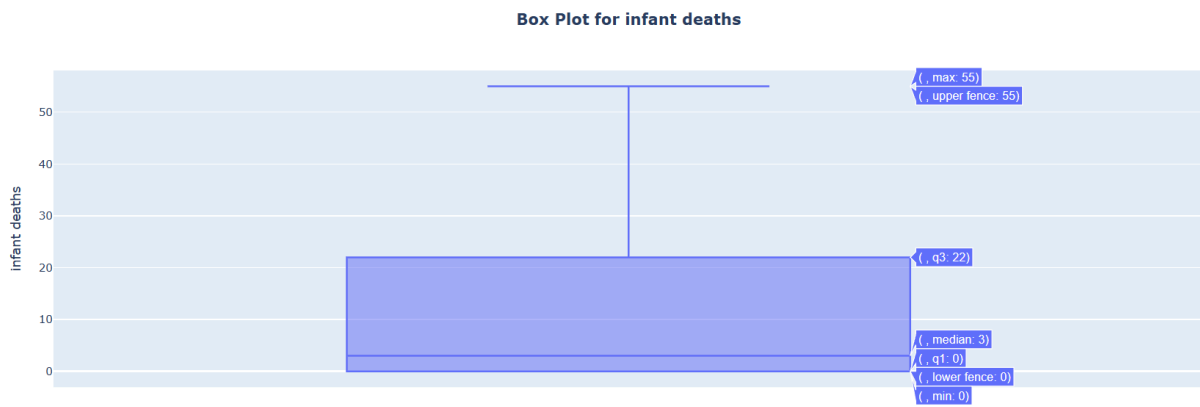


Fig. 31. Outlier removed from infant deaths

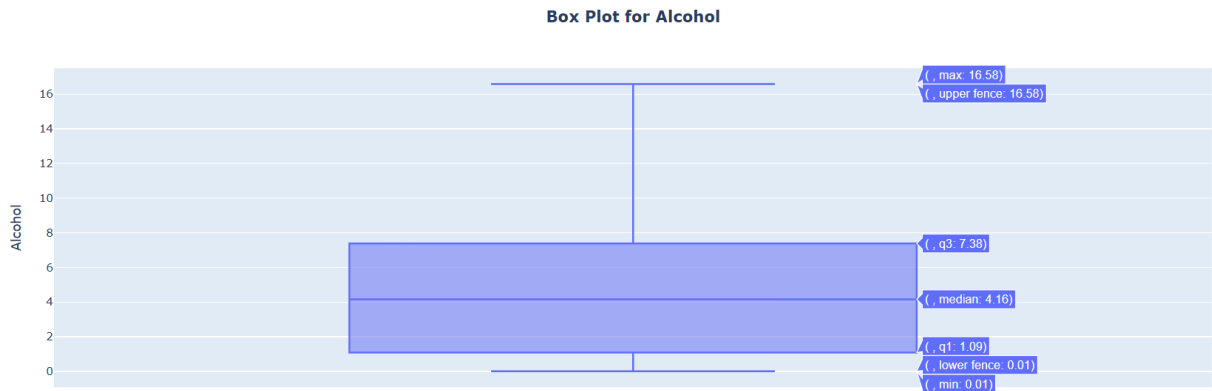


Fig. 32. Outlier removed from Alcohol

6.3 Encoding of Categorical Feature

As In our dataset we have two categorical feature 'Country' and 'Status'. So, In last step of preprocessing we will encode all the categorical features in numerical value. So we can directly give our data to Regression model. By using Label encoding for 'Country' and 'Status' to encode in numerical values. The Result can be checked in Figure 33.


```
df['Country'].unique()
```

```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12,
       13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25,
       26, 44, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37,
       38, 39, 40, 41, 42, 43, 45, 46, 47, 48, 49, 50, 51,
       52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64,
       65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77,
       78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90,
       91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103,
      104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116,
      117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129,
      130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142,
      143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155,
      156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168,
      169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181,
      182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192])
```

```
df['Status'].unique()
```

```
array([1, 0])
```

Fig. 33. Label encoding on Categorical features

7. Advanced EDA after Preprocessing (EDA Second Phase)

- To perform Advance EDA after preprocessing we are using plotly.express which specified library to generate interactive histograms, animations, and 3D scatter plots to deepen insights post-preprocessing. To check these visualization and insights, refer to Figure(34 to 43).

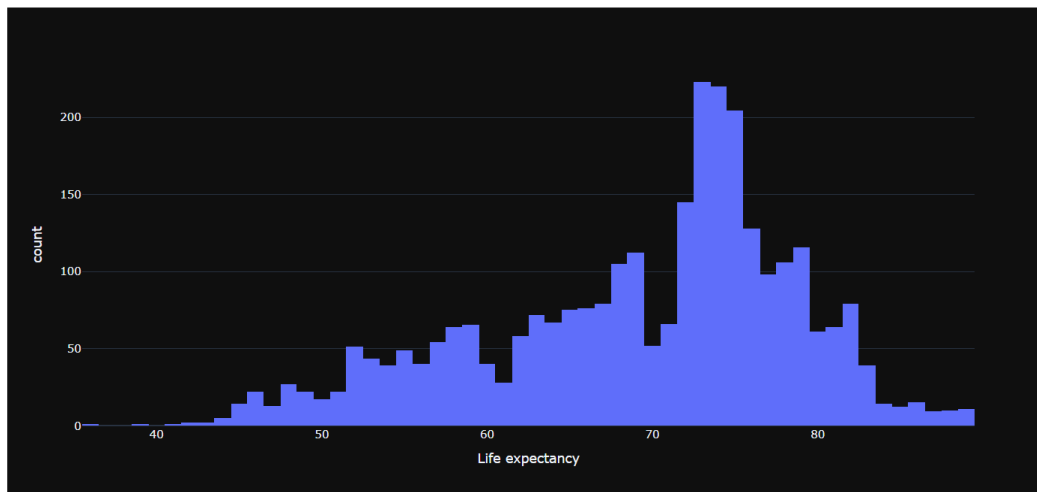


Fig. 34 Histogram Graph Between 'Life Expectancy' and 'Count'

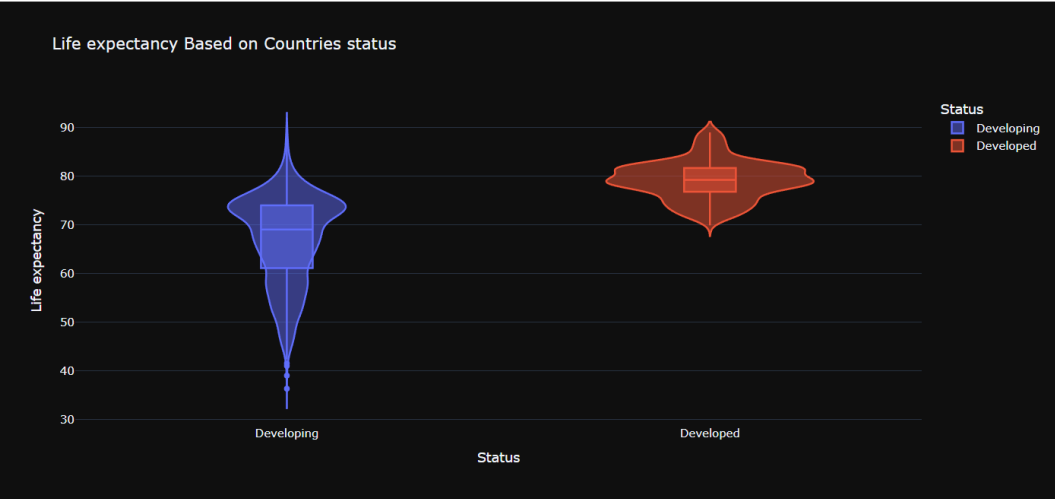


Fig. 35. Violin Graph shows 'Life Expectancy' based on the country's status

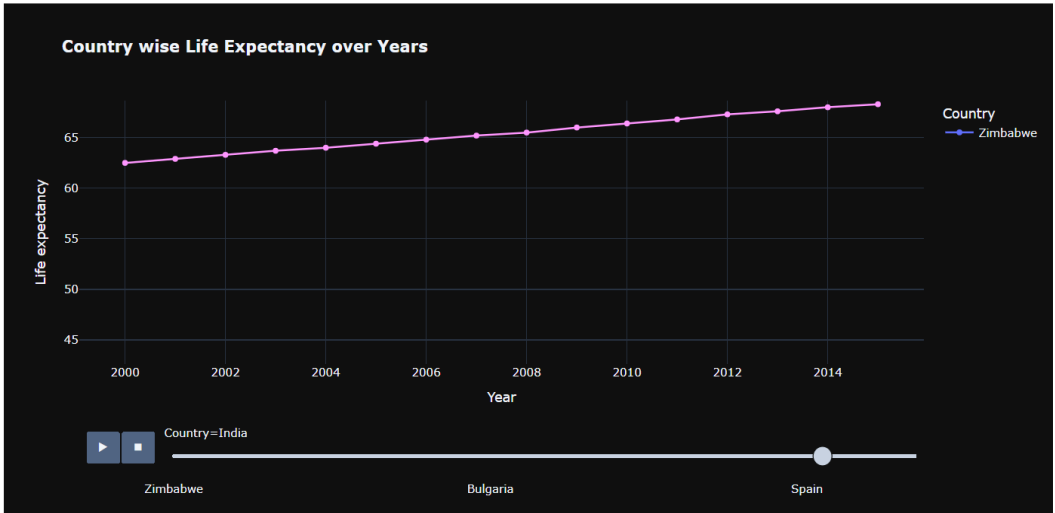


Fig. 36 Animation on 'Life Expectancy' over Years of all Countries



Fig. 37. Scatterplot Graph between 'Life Expectancy' and Adult Mortality of all countries

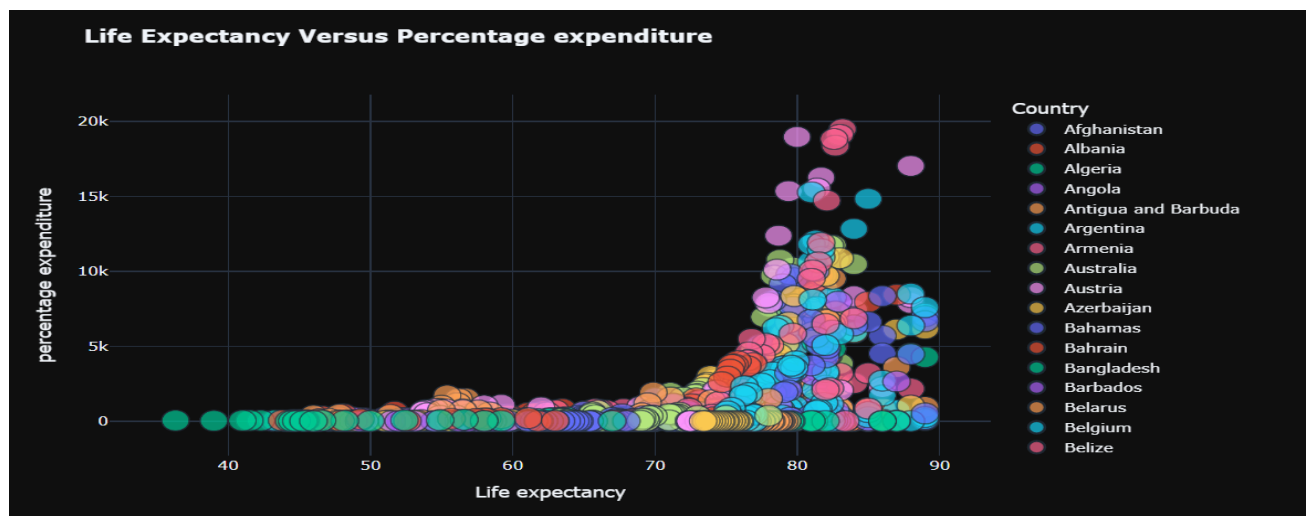


Fig. 38. Scatterplot Graph between 'Life Expectancy' and Percentage expenditure of all countries

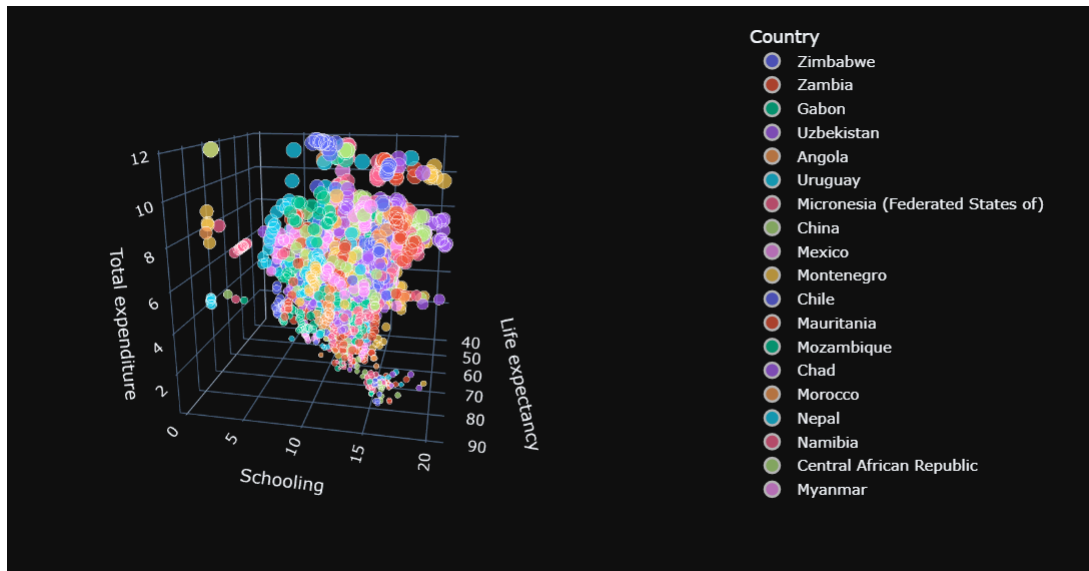


Fig. 39. 3D Scatterplot Graph between 'Life Expectancy', 'Schooling' and 'Total expenditure' of all countries

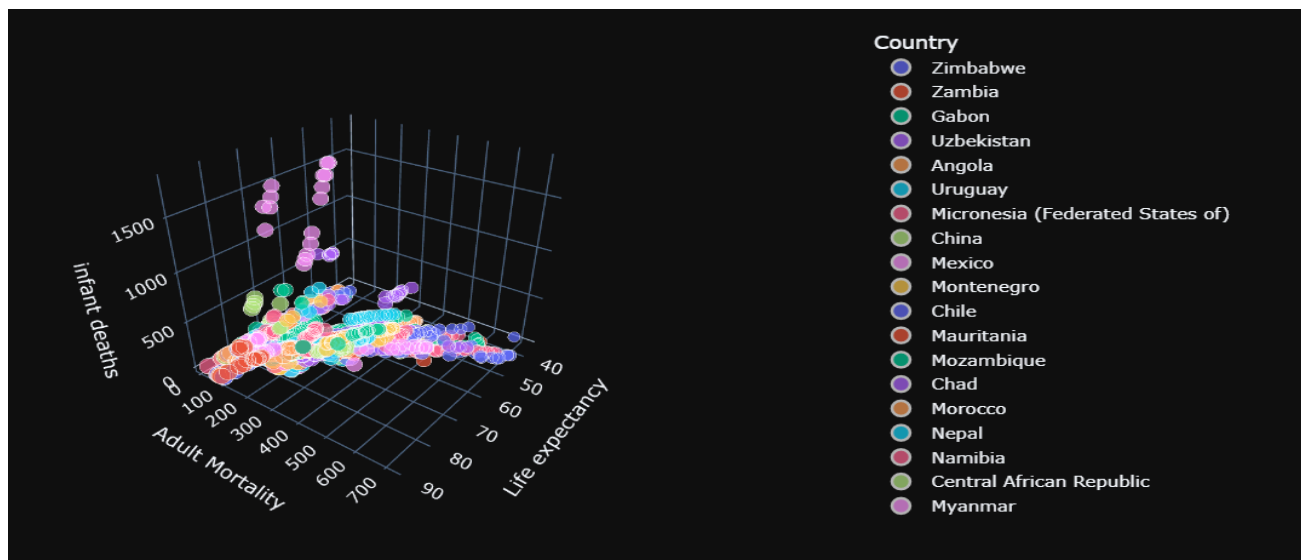


Fig. 40. 3D Scatterplot Graph between 'Life Expectancy', 'infant deaths' and 'Adult Mortality' of all countries

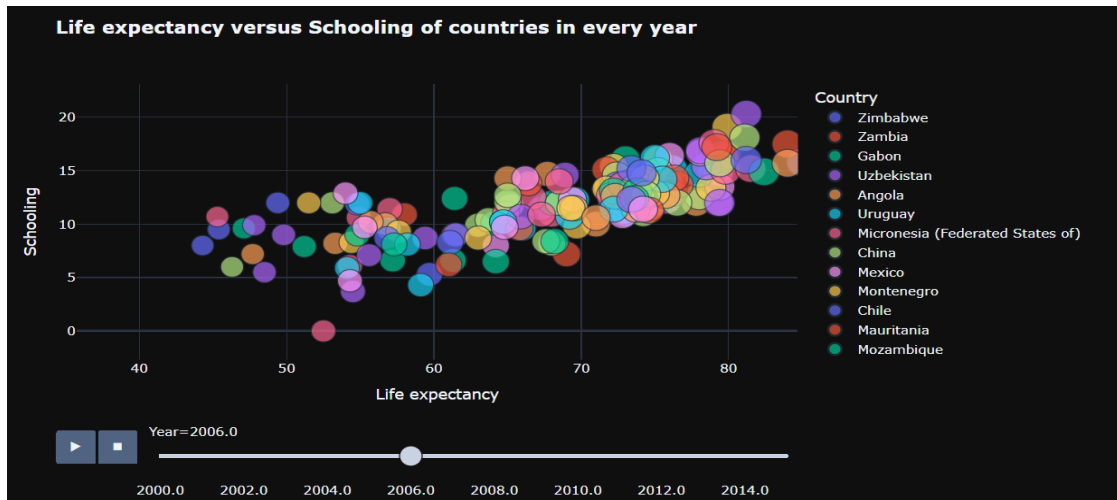


Fig. 41. Animation between 'Life Expectancy' and 'Schooling' over Years of all Countries

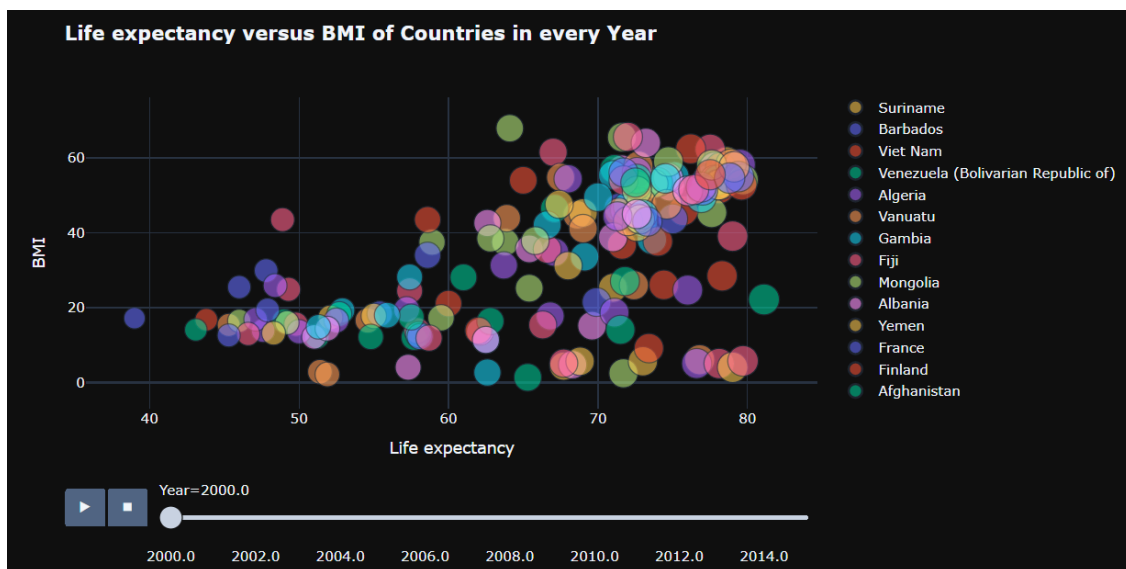


Fig. 42. Animation between 'Life Expectancy' and 'BMI' over Years of all Countries

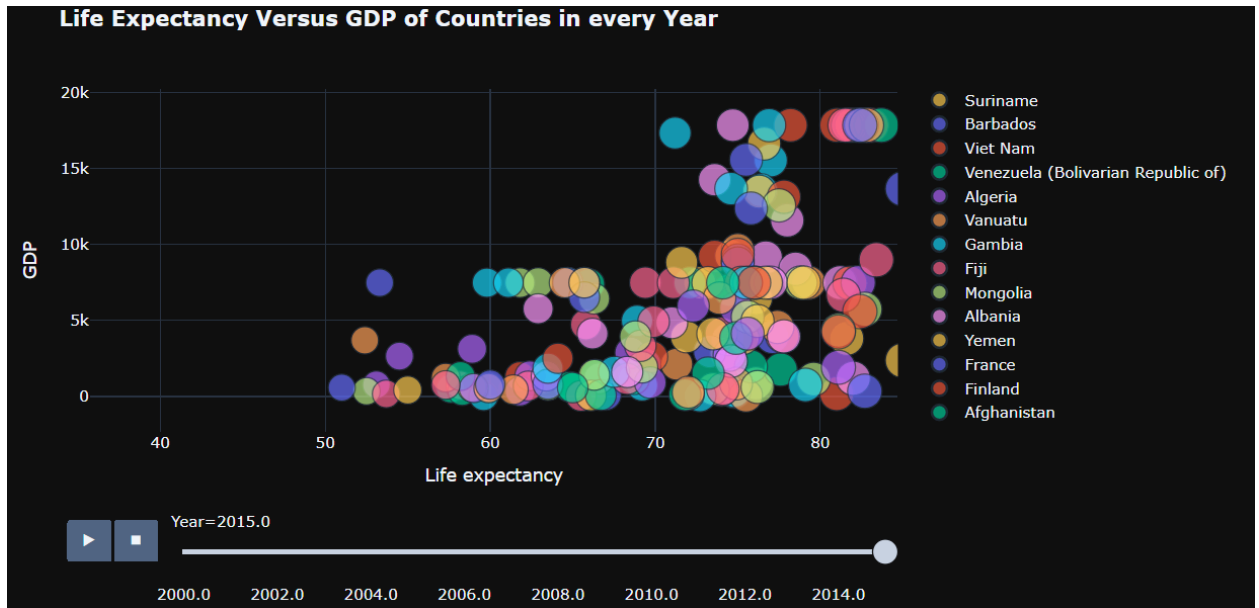


Fig. 43. Animation on 'Life Expectancy' and 'GDP' over Years of all Countries

8. Lazy Predict (Model Selection and Evaluation)

- The LazyPredict library was employed to see fast benchmark of all multiple regression models ([9]), providing metrics such as Adjusted R-Squared, R-Squared, RMSE, and computation time in Figure 44. This help us to identified best Regression models to predict Life Expectancy.

• The Results are:-

Model	Adjusted R-Squared	R-Squared	RMSE	Time Taken
ExtraTreesRegressor	0.97	0.97	1.50	2.78
HistGradientBoostingRegressor	0.97	0.97	1.62	0.81
LGBMRegressor	0.97	0.97	1.69	0.38
XGBRegressor	0.97	0.97	1.71	2.39
RandomForestRegressor	0.96	0.97	1.73	8.69
BaggingRegressor	0.95	0.96	1.96	0.70
GradientBoostingRegressor	0.94	0.95	2.18	3.62
KNeighborsRegressor	0.92	0.92	2.55	0.05
DecisionTreeRegressor	0.92	0.92	2.64	0.15
SVR	0.90	0.90	2.95	1.23
NuSVR	0.89	0.90	3.00	0.85
AdaBoostRegressor	0.89	0.90	3.00	0.91
ExtraTreeRegressor	0.89	0.89	3.07	0.10
HuberRegressor	0.85	0.86	3.54	0.07
ElasticNetCV	0.85	0.86	3.54	0.43
SGDRegressor	0.85	0.86	3.54	0.05
RidgeCV	0.85	0.85	3.55	0.14
BayesianRidge	0.85	0.85	3.55	0.02
LassoCV	0.85	0.85	3.55	0.94
LassoLarsCV	0.85	0.85	3.55	1.14
LarsCV	0.85	0.85	3.55	0.50
Ridge	0.85	0.85	3.55	0.15
TransformedTargetRegressor	0.85	0.85	3.55	0.17
LinearRegression	0.85	0.85	3.55	0.36
LassoLarsIC	0.85	0.85	3.55	0.21
Lars	0.85	0.85	3.55	0.23
LinearSVR	0.85	0.85	3.55	0.34
PoissonRegressor	0.85	0.85	3.56	0.04
OrthogonalMatchingPursuitCV	0.84	0.84	3.71	0.04
ElasticNet	0.82	0.83	3.86	0.05
Lasso	0.82	0.82	3.92	0.13
LassoLars	0.82	0.82	3.92	0.20

Fig. 44. Result of Lazy Predict

9. Enhancing the performance of Models by Hyperparameters.

From seeing lazy predict we choose our 9 Models shown in Figure 45 on the basis of their performance and diversity in their mathematical formula we choose them for Hyperparameter-Tuning to enhance their R2 score and minimize their RMSE. We use GridSearchCV to tune Hyperparameter of all 9 models shown in Figure 46.

```
models = {
    "LinearRegression": (LinearRegression(), StandardScaler()),
    "Ridge": (Ridge(alpha=1.0), StandardScaler()),
    "Lasso": (Lasso(alpha=0.1), StandardScaler()),
    "SVR": (SVR(kernel='rbf'), StandardScaler()),
    "DecisionTree": (DecisionTreeRegressor(), MinMaxScaler()),
    "RandomForest": (RandomForestRegressor(), MinMaxScaler()),
    "GradientBoosting": (GradientBoostingRegressor(), MinMaxScaler()),
    "KNN": (KNeighborsRegressor(n_neighbors=5), StandardScaler()),
    "XGBoost": (XGBRegressor(), MinMaxScaler())
}
```

Fig. 45. Models for Hyperparameter tuning w.r.t Normalization

```
param_grid = {
    "LinearRegression": {},
    "Ridge": {'model__alpha': [0.01, 0.1, 1, 10, 100]}, # Regularization strength
    "Lasso": {'model__alpha': [0.001, 0.01, 0.1, 1, 10]}, # Regularization strength
    "SVR": {
        'model__C': [0.1, 1, 10, 100], # Regularization parameter
        'model__epsilon': [0.01, 0.1, 0.2] # Defines tolerance margin for regression
    },
    "DecisionTree": {
        'model__max_depth': [None, 5, 10, 20], # Limits overfitting
        'model__min_samples_split': [2, 5, 10] # Minimum samples required to split a node
    },
    "RandomForest": {
        'model__n_estimators': [50, 100, 200], # Number of trees
        'model__max_depth': [None, 5, 10], # Limits overfitting
        'model__min_samples_split': [2, 5, 10] # Controls when a node should split
    },
    "GradientBoosting": {
        'model__n_estimators': [50, 100, 200], # Number of boosting stages
        'model__learning_rate': [0.01, 0.1, 0.2], # Shrinks contribution of each tree
        'model__max_depth': [3, 5, 10] # Depth of trees
    },
    "KNN": {'model__n_neighbors': [3, 5, 7, 10], # Number of neighbors
            'model__p': [1, 2] # Distance metric: 1 for Manhattan, 2 for Euclidean
    },
    "XGBoost": {
        'model__n_estimators': [50, 100, 200], # Number of boosting rounds
        'model__learning_rate': [0.01, 0.1, 0.2], # Step size shrinkage
        'model__max_depth': [3, 5, 10] # Depth of trees
    }
}
```

Fig. 46. GridSearchCV to tune Hyperparameter

10. Result and Discussion

The performance of Models after enhancing their performance by Hyperparameters tuning are shown in Figure 47. From These results, We can see that XGBRegressor out performs the best with 96.87% (R2 Score) shown in Figure 48.

	Model	Best Params	MAE	MSE	RMSE	R ²
0	LinearRegression	{}	2.68	12.59	3.55	0.85
1	Ridge	{'model__alpha': 10}	2.67	12.57	3.55	0.85
2	Lasso	{'model__alpha': 0.001}	2.68	12.59	3.55	0.85
3	SVR	{'model__C': 100, 'model__epsilon': 0.1}	1.41	4.84	2.20	0.94
4	DecisionTree	{'model__max_depth': 10, 'model__min_samples_s...	1.63	6.32	2.51	0.93
5	RandomForest	{'model__max_depth': None, 'model__min_samples...	1.08	2.93	1.71	0.97
6	GradientBoosting	{'model__learning_rate': 0.1, 'model__max_dept...	1.09	2.74	1.66	0.97
7	KNN	{'model__n_neighbors': 3, 'model__p': 1}	1.22	3.72	1.93	0.96
8	XGBoost	{'model__learning_rate': 0.2, 'model__max_dept...	1.08	2.71	1.65	0.97

Fig. 47. Regression Model metrics after Hyperparameter Tunning

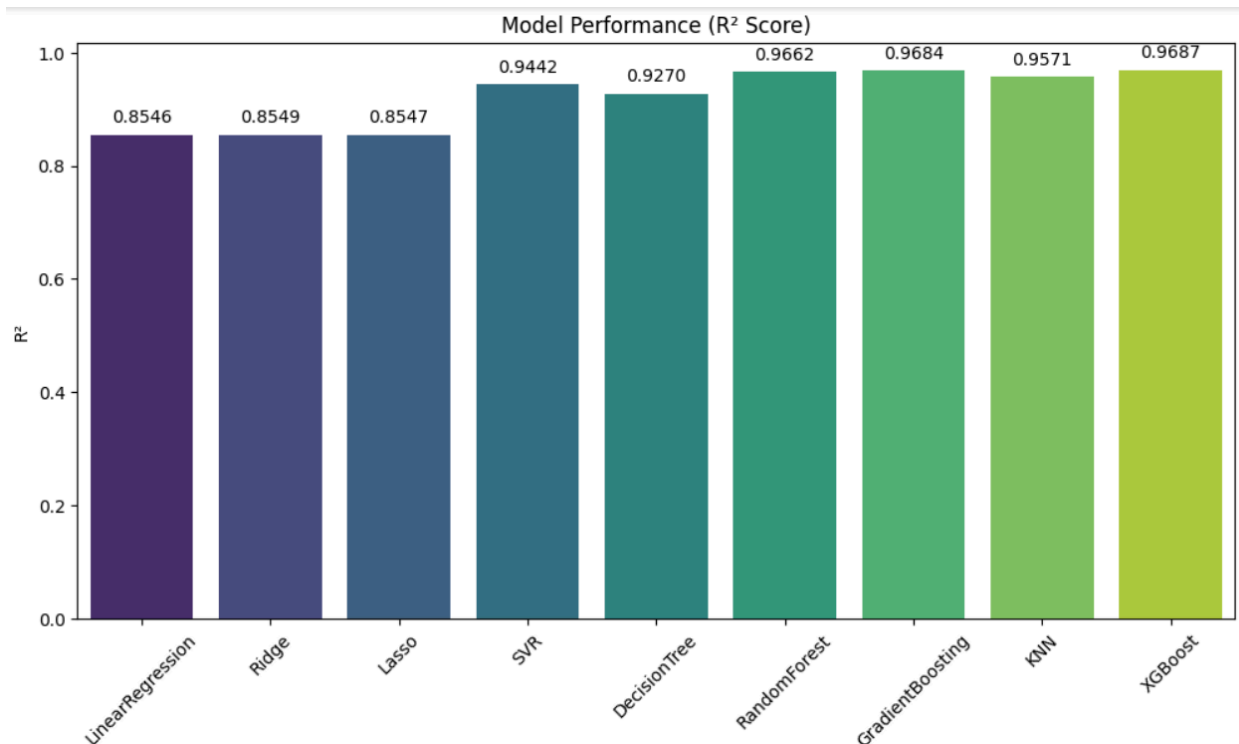


Fig. 48. The Graphical Representation of Model Performance

- As in previous result we see XGBoost outperform so validating it by performing Cross Validation score which is turn to be 95.54% shown in the Figure 51.

Cross-Validation R² Scores: [0.95710007 0.96618211 0.95506959 0.9494274 0.94964199]
Mean Cross-Validation R² Score: 0.9554842309550959
Mean Squared Error: 2.9090252465640924
R-squared Score: 0.9664221625369424

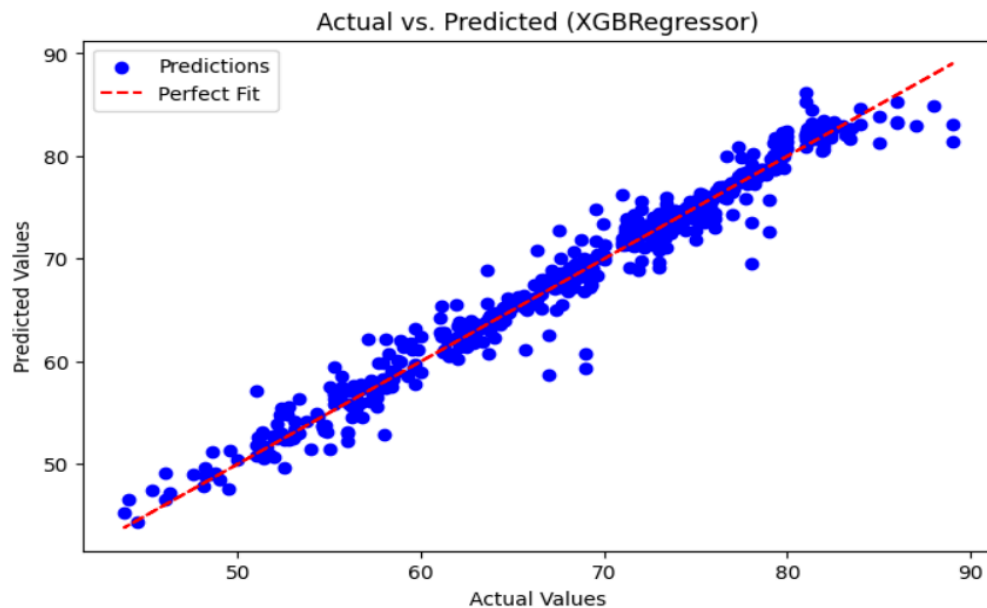


Fig. 49. Cross Val score with Best-Fit Line of XGBRegressor

- The results of 10 random value were also shown with their predicted values to compare the difference what model predicting Life Expectancy and actual Life Expectancy is shown in Figure 50.

Random Actual vs Predicted values:

	Actual	Predicted
0	67.60	67.82
1	86.00	83.37
2	73.80	73.39
3	67.00	62.50
4	70.00	71.34
5	71.50	71.32
6	74.40	72.78
7	77.30	80.95
8	81.60	82.92
9	81.00	82.11

Fig. 50. Shown the difference between Actual and Predicted values

11. Conclusion: -

In this paper, we collected a Kaggle dataset (original resource is WHO) with 2,938 samples from 193 countries with several features. The dataset contains 21 features with 'Life Expectancy' as the target variable. We visualize the various relationships between independent variables and dependent variable. After preprocessing the data, we applied lazy predict for fast benchmark score on all regression algorithms. From seeing lazy predict we choose our 9 algorithm models namely Multiple Linear Regression, Ridge, Lasso, Support Vector Regression (SVR), Decision Tree Regression, Random Forest, Gradient Boosting, KNN, XGBRegressor (XGBoost) on the basis of their performance and diversity in their mathematical formula we choose them for Hyperparameter-Tuning to enhance their R2 score and minimize their RMSE. After applying Hyperparameter-Tuning, we achieved best result with XGBRegressor (XGBoost) shown in Figure 48 and Figure 49.

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